Multidimensional Real Analysis - Duistermaat, Kolk

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October 21, 2026

1. Chapter 1 - Continuity
   (a) section 1.1 - Inner product and Norm
   - (Defn) Inner product and Norm
   - (Thm) Properties of Inner Product
   - (Thm) Cauchy-Schwarz Inequality
     Proof Idea: Decompose a vector x as its projection onto y plus its orthogonal component. Take the norm squared and solve for the orthogonal component. Its nonegegativity forces the CS inequality.
   - (Thm) Inequalities related to the norm
     - Triangle inequality and extended triangle inequality
     - Reverse triangle inequality
     - $|x_j| \leq ||x|| \leq \sum_{i=1}^{n} |x_i| \leq \sqrt{n}||x||$
       Proof Idea: First ineqality is straight forward, second inequality can be seen by decomposing the vector and applying the triangle inequality. Third inequality can be seend by forming the vector $x_{\text{abs}}|$ composed of the absolute values. Then take the inner product with the vector of all 1s and apply C.S.
     - $\max_j |x_j| \leq ||x|| \leq \sqrt{n} \max_j |x_j|$. This implies that for a cube or side length 2 is enclosed in a sphere of radius $\sqrt{n}$ which is enclosed in a cube of side length $2\sqrt{n}$.
       Proof Idea: First inequality follows from the previous part. 2nd inequality can be seen by replacing each component of x with the maximum component. The norm of the new vector is the RHS.
   - (Defn) Distance between two vectors
   - (Defn) Convergence or limit of a sequence of vectors in $\mathbb{R}^n$.
   - (Thm) A sequence of vectors converges iff each sequence of components converges.
   - (Thm) The limit operation on vectors ist still linear.

   (b) Section 1.2 - Open and closed sets
   - (Defn) Open Ball/neighborhood
   - (Thm) Arbitrary intersections of open sets are open. FINITE intersections of open sets are open. Reverse is true for close sets.

- (Thm) Closed ball is a closed set.
- (Defn) Cluster Point of a set, Closure of a set
- (Thm) $(\bar{A})^c = \text{int}(A^c)$
  Note that in the algebra of closure and interior operators, a complement (on the outside, swaps interior for closure and moves to the inside.
- (Thm) Equivalent definitions of closed.
  - F is closed.
  - $F = \bar{F}$
  - Every convergent sequence of points contained in F converges to a point IN F.

(c) Section 1.3 - Limits and Continuous Mappings

- (Defn) jth partial mapping
- (Defn) Limit of a function at a point.
  Note the rephrasing in terms of epsilon and delta balls.
- (Thm) The limit exists if and only if the preimage of every neighborhood is a neighborhood. (recall a neighborhood is just a set that contains an open ball containing the point).
- (Thm) The limit exists iff every sequence that converges to a, the image of that sequence of that sequence converges to b.